Predicting Unemployment through Trade Networks: A Factor Network Autoregression Approach

Giacomo Rinaldi (0001176098)
Matteo Severi (0001174713)
Gianluca Antonio Spennacchio (0001183602)

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1 Introduction

Modern economies are organised in complex networks where countries and sectors interact through multiple layers, such as trade, production, and finance. When macroeconomic variables are jointly driven by these linkages, models that neglect the network structure risk misrepresenting the propagation of shocks and the cross-sectional dependence that characterises global systems.

The Factor Network Autoregression (FNAR) introduced by Barigozzi et al. (2025) provides a tractable framework to model such interdependence in a high-dimensional setting. The model starts from a multilayer network representation, where each layer corresponds to a distinct type of bilateral relation. It assumes that the observed layers $W_{k,t}$, $k = 1, \dots, m$, share a common factor structure, so that the evolution of the entire multilayer network can be represented by a small number of latent network factors $F_{j,t}$, each taking the form of an $N \times N$ matrix.

These latent components capture the cross-layer dependence and, if sufficiently pervasive, allow us to replace the original multilayer specification with a FNAR, yielding a model with r network factors, where r denotes the number of common network factors (typically $r \ll m$ and treated as fixed with respect to N and m). The FNAR takes the form of a VAR(1) augmented with a network-mediated term constructed from the lagged factors, so that

$$y_t = \alpha + \rho\, y_{t-1} + \sum_{j=1}^{r} \beta_j \left(\frac{F_{j,t-1}}{N}\right) y_{t-1} + \nu_t. \tag{1}$$

Here α, ρ and β_j are scalars, ν_t is the N-dimensional vector of FNAR errors. In practice, estimation proceeds by replacing $F_{j,t}$ with their empirical counterparts $\widehat{F}_{j,t}$ extracted from the layers. The rescaling by N is consistent with the row normalisation of the observed networks, which we will further discuss in Subsection 3.1.

From an econometric standpoint, the FNAR can be interpreted as a static approximate factor model for network-valued data. The static nature follows from the estimation of factors using only contemporaneous sample second moments of the multilayer network, while the approximate component allows the idiosyncratic tensor to exhibit weak cross-sectional and temporal dependence. Identification relies on the cross-sectional variation of the network, analogously to standard principal component analysis, which provides consistent estimation of the space spanned by the common factors even in large dimensions.

The empirical application investigates whether trade-based network factors contain predictive information for unemployment dynamics. Using data from the World Input-Output Tables (WIOT), we construct a `Country` $\times$ `Country` $\times$ `Sector` $\times$ `Time` tensor describing the evolution of international trade networks. After cleaning and harmonising bilateral flows (see Subsection 3.1) we extract network factors through tensor-based principal component analysis. These estimated factors are then used within the FNAR framework to assess their role in forecasting changes in national unemployment rates, expressed in first differences to ensure stationarity.

In particular, we set out a first estimation based on all data available, to study the linkage between the factors and unemployment rate changes from 2001 to 2014.
However, we also perform two estimations on the data dividing the sample in pre-2008 years and post-2008 years. We do this to investigate whether the relationship between the identified factors and unemployment rate change vary after the Financial and Sovereign Debt crises that characterised the second set of years.
Our findings suggest that, indeed, there is a radical modification happening from the first period to the second one, with factors showing significant coefficients only in the latter time frame.

The remainder of the paper is organised as follows. Section 2 presents the FNAR framework and its estimation procedure. Section 3 describes the empirical application, including data treatment, factor extraction, and estimation results. Section 4 concludes.

Research Question: Do sectoral trade interdependencies captured in WIOT data contain predictive information about the evolution of unemployment across countries?

2 FNAR framework

2.1 Model

Let $\mathcal{W}_t \in \mathbb{R}^{N \times N \times m}$ collect the m network layers at time t. We assume

$$\mathcal{W}_t \;=\; \mathcal{F}_t \times_3 U \;+\; \mathcal{E}_t, \qquad t = 1, \ldots, T, \tag{2}$$

where $\mathcal{F}_t \in \mathbb{R}^{N \times N \times r}$ has frontal slices $F_{j,t} \in \mathbb{R}^{N \times N}$, $U \in \mathbb{R}^{m \times r}$ are layer loadings, and $\mathcal{E}_t$ is idiosyncratic. Writing the mode-3 matricization $\mathcal{W}_t^{(3)} \in \mathbb{R}^{m \times N^2}$,

$$\mathcal{W}_t^{(3)} \;=\; U\,\mathcal{F}_t^{(3)} \;+\; \mathcal{E}_t^{(3)}, \qquad t = 1, \ldots, T. \tag{3}$$

The outcome $y_t \in \mathbb{R}^N$ follows the FNAR in Equation (1), i.e. a VAR(1) with the network-mediated term built from $(1/N)F_{t-1}y_{t-1}$ (Barigozzi et al., 2025, Sec. 3).

Node factors in errors. To allow for components common across nodes, let

$$\nu_{it} \;=\; \lambda_{i1}G_{1t} \;+\; \cdots \;+\; \lambda_{iq}G_{qt} \;+\; \epsilon_{it}, \qquad i=1,\dots,N, \quad t=1,\dots,T, \tag{4}$$

with G_{kt} node factors and λ_{ik} loadings. Vector form: $\nu_t = \Lambda G_t + \epsilon_t$ (Barigozzi et al., 2025, Sec. 3).

Decomposition of Equation (1)**.** The term $\sum_{j=1}^{r} \beta_j \big(\frac{F_{j,t-1}}{N}\big) y_{t-1}$ is the *network effect*; $\rho\, y_{t-1}$ is the *momentum effect*; α is the *nodal effect.*

The *network effect* represents dynamic spillovers across nodes mediated by the latent network factors. It captures how a change in one unit transmits to others through the multilayer structure (via factor loadings), allowing propagation to be time-varying and heterogeneous across destinations.
The *momentum effect* defines own-lag persistence at the node level. It isolates intrinsic inertia in each series that is not attributable to network transmission, i.e., the direct dependence of current outcomes on their past values.
The *nodal effect* corresponds to a common level component shared by all nodes. It absorbs shocks or slow-moving shifts that affect the cross-section uniformly and can be generalised to include common regressors (e.g., node-level covariates with a common parameter).

2.2 Estimation

Network factor extraction. Define the layers-mode second-moment matrix

$$\widehat{\Gamma}^{\mathcal{W}} \;=\; \frac{1}{T}\sum_{t=1}^{T} \mathcal{W}_t^{(3)}\, \mathcal{W}_t^{(3)\prime} \in \mathbb{R}^{m\times m}. \tag{5}$$

Let $\widehat{U}$ collect the r leading eigenvectors of $\widehat{\Gamma}^{\mathcal{W}}$; obtain $\widehat{\mathcal{F}}_t^{(3)}$ by projection and reshape to $\widehat{\mathcal{F}}_t$. Factors are estimated from contemporaneous second moments, allowing the idiosyncratic tensor to be autocorrelated (Barigozzi et al., 2025, Sec. 4).

Number of factors. Let $\widehat{\mu}_j^{\mathcal{W}}$ be the jth largest eigenvalue of $\widehat{\Gamma}_W$. Estimate r by the eigenvalue-ratio rule

$$\widehat{r} \;=\; \arg\max_{j=1,\dots,r_{\max}} \frac{\widehat{\mu}_j^{\mathcal{W}}}{\widehat{\mu}_{j+1}^{\mathcal{W}}}. \tag{6}$$

Let $\widehat{\mu}_j^{\nu}$ be the jth largest eigenvalue of the sample covariance of FNAR residuals; estimate node factors q by

$$\widehat{q} \;=\; \arg\max_{j=1,\dots,q_{\max}} \frac{\widehat{\mu}_j^{\nu}}{\widehat{\mu}_{j+1}^{\nu}}, \tag{7}$$

with the standard iterative update on residuals (Barigozzi et al., 2025, Sec. 4).

FNAR coefficients. Let $\widehat{X}_t \;:=\; \big[\,(1/N)\,\widehat{F}_{1,t-1}y_{t-1}, \quad \cdots \quad (1/N)\,\widehat{F}_{r,t-1}y_{t-1}, \quad y_{t-1}, \quad \iota_N\,\big] \in \mathbb{R}^{N\times(r+2)}$, where ι_N is a N-dimensional vector of ones, and $\widehat{F}_{j,t-1} \in \mathbb{R}^{N\times N}$ the j-th estimated network factor matrix at time t-1. Let $\theta := (\beta_1, \ldots, \beta_r,\ \rho,\ \alpha)'$.

$$\widehat{\theta}_{\text{OLS}} = \Big(\sum_{t=1}^{T} \widehat{X}_t' \widehat{X}_t\Big)^{-1} \Big(\sum_{t=1}^{T} \widehat{X}_t' y_t\Big). \tag{8}$$

Let $V := \text{Var}(\nu_t) \in \mathbb{R}^{N \times N}$ denote the contemporaneous covariance matrix of FNAR errors, and let $\widehat{V}$ be a consistent estimator of V. The GLS-type estimator uses a consistent $\widehat{V}^{-1}$:

$$\widehat{\theta}_{\text{GLS}} \ = \ \Big(\frac{1}{NT} \sum_{t=1}^{T} \widehat{X}_t' \widehat{V}^{-1} \widehat{X}_t\Big)^{-1} \Big(\frac{1}{NT} \sum_{t=1}^{T} \widehat{X}_t' \widehat{V}^{-1} y_t\Big). \tag{9}$$

See Barigozzi et al. (2025, Sec. 4, Sec. 5 and App. B–C) for further details and specifications.

2.3 Assumptions and large-sample properties

Assume: (i) r is finite and independent of N and m; (ii) weak dependence and moment bounds for $\mathcal{E}_t$ allowing autocorrelation over time; (iii) pervasiveness across layers; (iv) standard regularity for U and factor processes; (v) $(T, m, N) \to \infty$ under the growth conditions in Barigozzi et al. (2025, Sec. 5). Under these, the FNAR has a weakly stationary and causal solution and estimators are consistent and asymptotically normal. In particular, for GLS,

$$\sqrt{NT}\,(\widehat{\theta}_{\text{GLS}} - \bar{J}\theta) \ \xrightarrow{d} \ \mathcal{N}(0_{r+2}, \Omega_1^{-1}), \tag{10}$$

with scaling requirements $\sqrt{NT}/m \to 0$ and $N/m \to 0$; OLS entails weaker scaling. See Barigozzi et al. (2025, Sec. 4, Sec. 5 and App. B–C) for further details and specifications.

3 Empirical Application

3.1 Data

The data we use in our empirical application of the methodology described so far come from World Input Output Tables. This data set reports trade between a total of 42 countries from 2000 to 2014 across 56 different industries belonging to the primary, secondary and tertiary sector. The countries reported in the dataset are mostly located in Europe (slightly less than 60% of the total sample), with some other advanced and developing economies also being represented (like the USA, China, Japan, South Korea, Japan, Indonesia).
The data contain reference to Taiwan, Rest of the World and a Total tracker of trades across countries; however, these are not retained in our analysis as they lack a counterpart in World Bank datasets for unemployment (being our final dependent variable).

Before proceeding towards cleaning, harmonising and shaping the data in the right format, we check the availability of data and drop sector 56 (*Activities of extraterritorial organizations and bodies*). Indeed, all the observations associated to it are 0, suggesting that its presence would not be relevant to our case.

Then, the main steps we carry out are about treating missing information, which is mostly presented in the form of all-zero rows across the dataset.

For each panel unit i and year t, missing observations are imputed using a leave–one–out moving average over the two adjacent years on each side. Define $\mathcal{K} = \{-2, -1, 1, 2\}$ and the indicator $\mathbb{1}_{i,t+k} = \mathbb{1}\{y_{i,t+k} \text{ is observed}\}$. Then,

$$\tilde{y}_{i,t} = \begin{cases} \dfrac{\sum_{k\in\mathcal{K}} \mathbb{1}_{i,t+k}\, y_{i,t+k}}{\sum_{k\in\mathcal{K}} \mathbb{1}_{i,t+k}}, & \text{if } \sum_{k\in\mathcal{K}} \mathbb{1}_{i,t+k} \geq 2, \\ \text{missing}, & \text{if } \sum_{k\in\mathcal{K}} \mathbb{1}_{i,t+k} = 0. \end{cases}$$

The target year t is excluded (leave–one–out), and only available neighboring observations within the same panel i are used. For boundary years, the mean is computed over the existing adjacent observations only.
It should be noted that, before using these averages to impute the values, we verify that standard deviation between available values and the imputed ones is not too large (which would suggest that the data are too volatile to predict their behavior). The applied method allows us to impute a little more than 1,600 entries, that had previously been registered as zeros.
After this, we compute the weights for the entries of the tensor, as reported in the paper from Barigozzi et al. (2025, App. G). The formula is

$$\mathcal{W}_{ijk,t} = N \frac{\text{imports}_{ij,k,t} + \text{exports}_{ij,k,t}}{\sum_{h=1}^{N} \left(\text{imports}_{ih,k,t} + \text{exports}_{ih,k,t}\right)}$$

where $\text{imports}_{ij,k,t}$ represents all imports of good k at time t to country i from country j, while $\text{exports}_{ij,k,t}$ represents all exports of the same good from country i to country j.

Once this normalisation is computed, we again analyse the obtained data set to check for possible other sectors or countries that report suspicious data information. We find that indeed Russia presents only-zero-entry rows in many of the sectors in the dataset, including some where the country is actually known to have trade (see fishing and aquaculture, for example). For this reason, we decide to drop the country due to the unreliability of the data.

After this, we proceed with the analysis of sectors, which also present some issues with respect to all-zero rows; consequently we decide to drop an additional 10 out of the 55 remained fields in the data set. This is done mainly due to the lack of information on many countries. Table 3 in the Appendix shows a complete list of all the sectors dropped (including number 56 which was removed from the beginning).
As it can be noticed from the description, these sectors are either quite niche, or are likely to have been affected by some measurement error, as the value of the trade might have been hard to track.

Although we have lost some information, we still remain with a total of 45 sectors, which we deem to be well enough to proceed with our analysis.
Consequently, we recompute the imputation and standardisation steps, excluding both the sectors and country we identified, and obtain the needed data set.
Finally, we use external data from the World Bank to fill some missing entries that still appear in the matrix. In doing so, we avoid obtaining unreliable results from our analysis.

In Table 4 and Table 5 in the Appendix are provided the lists of the countries still present in the dataset and the sectors retained.

On the other hand, we use World Bank, 2025 data sets for information on unemployment rates during the time considered in WIOT. To be able to use this data, however, we need to take first differences of the shares reported in the data set. In fact, this transformation enables us to consider the dependent variable as stationary, which is needed for the theoretical assumptions of the model to hold.

3.2 Tensor-based PCA

This subsection outlines the factor-extraction step for multilayer networks by exploiting cross-sectional variation across layers, in direct analogy with ordinary PCA, and without using temporal dynamics.

Starting from the layers–mode representation in Equation (3), compute the contemporaneous second–moment matrix across layers, Equation (5). Let $\widehat{V}^{\mathcal{W}} \in \mathbb{R}^{m\times r}$ collect the r normalized eigenvectors associated with the r largest eigenvalues of $\widehat{\Gamma}^{\mathcal{W}}$, and let $M^{\mathcal{W}} \in \mathbb{R}^{r\times r}$ be the diagonal matrix with those eigenvalues on the diagonal. The layers loading matrix is then estimated as

$$\widehat{U} \;:=\; \frac{1}{N}\,\widehat{V}^{\mathcal{W}}\,(\widehat{M}^{\mathcal{W}})^{1/2}$$

Given $\widehat{U}$, the mode-3 matricization of the network factors is obtained by linear projection of $\mathcal{W}_t^{(3)}$ onto span$(\widehat{U})$:

$$\widehat{\mathcal{F}}_t^{(3)} \;:=\; (\widehat{U}'\widehat{U})^{-1}\widehat{U}'\,\mathcal{W}_t^{(3)} \;=\; N\,(\widehat{M}^{\mathcal{W}})^{-1/2}\,(\widehat{V}^{\mathcal{W}})'\,\mathcal{W}_t^{(3)},$$

and folding back gives the estimated $N \times N \times r$ tensor of network factors with frontal slices $\widehat{F}_{k,t}$, $k = 1, ..., r$ (Barigozzi et al., 2025, Sec. 4).

Because estimation uses only contemporaneous second moments, the idiosyncratic tensor can be autocorrelated without affecting the factor-extraction step.

As in standard PCA, factors are recovered from the eigenvectors of a sample covariance matrix; identification is up to rotation and fixed by choosing an orthonormal basis of leading eigenvectors; and consistency relies on pervasiveness of the common component and weak idiosyncratic dependence. The only difference is the mode along which covariance is computed (here, across layers), which is natural in multilayer networks (Barigozzi et al., 2025, Sec. 3, Sec. 4).

3.2.1 Number of Network Factors

Determining the dimension of the factor space is intrinsically delicate and rarely admits a single, definitive rule. Accordingly, we combine complementary formal and informal diagnostics to guide our choice. Formally, we employ the eigenvalue–ratio criterion to identify a dominant "gap" in the spectrum (Barigozzi et al., 2025). Informally, we examine a scree plot of variance shares and the growth of eigenvalues as the cross-sectional size N increases, as a pervasiveness check.

Intuitively, (6) identifies the index at which the spectrum exhibits the sharpest relative drop, signaling the transition from common to idiosyncratic components. In our application, the rule of the eigenvalue-ratio points to $\widehat{r} = 1$. In order to check whether our model admits more than one factor we proceed with the scree plot

The scree plot reports the fraction of total variance explained by each successive eigenvalue. A common heuristic is to identify the index beyond which the marginal contributions flatten, known as the *elbow*. Figure 1a suggests a plausible range of $r \in 2, 4$, as two elbows appear, one at the second eigenvalue and another at the fourth. Beyond the fourth eigenvalue, the additional variance explained by successive factors is minimal and no further sharp drops are observed.

Lastly we proceed with a check of the pervasiveness of the eigenvalues as the number of observations increases. Because our FNAR objects are tensors of size $N \times N \times m \times T$, increasing the cross-sectional size expands the number of network entries along the two country dimensions. Increasing only across one country dimension wouldn't respect the theory behind FNAR. Consequently, pervasiveness for truly common components should manifest as eigenvalues that diverge at a linear rate N^2 (with m fixed), rather than the linear-in-N growth typical of one-dimensional cross-sectional panels.

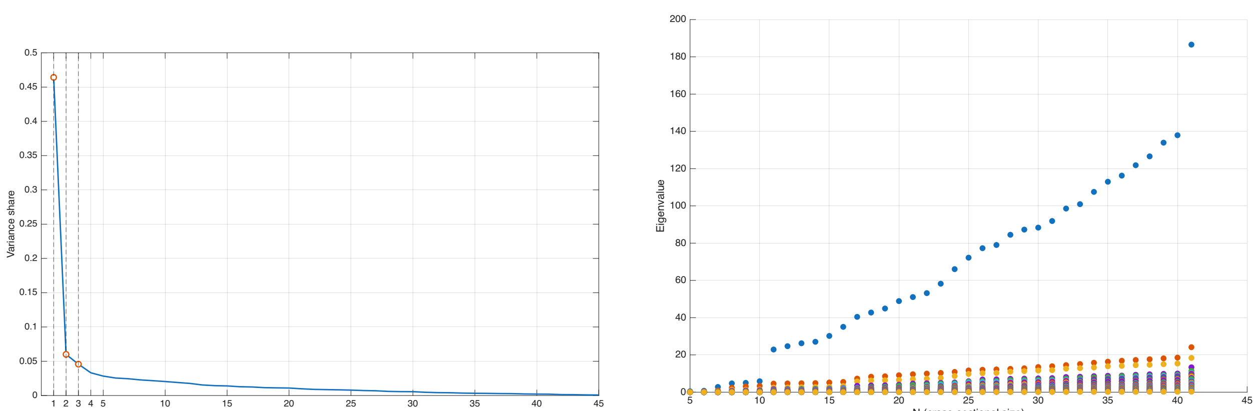
This property (see Appendix E of Barigozzi et al. (2025)) can be written as

$$\text{(i)} \quad \forall j = 1, \dots, r: \qquad \underline{C}_j \;<\; \liminf_{m,N\to\infty} \frac{\mu_j^X}{mN^2} \;\leq\; \limsup_{m,N\to\infty} \frac{\mu_j^X}{mN^2} \;<\; \overline{C}_j, \tag{11}$$

$$\text{(iv)} \quad \text{Let } \mu_1^\varepsilon \text{ be the largest eigenvalue of } \Gamma^\varepsilon. \text{ Then } \exists\, \gamma \in [0,2],\ M_\varepsilon < \infty \text{ such that } \frac{\mu_1^\varepsilon}{N^\gamma} \leq M_\varepsilon \tag{12}$$

Condition (11) formalises that network eigenvalues diverge at order mN^2 (bounded above and below by constants independent of m, N), while (12) restricts the idiosyncratic elements to grow no faster than N^γ with $\gamma \leq 2$.

Figure 1b plots the estimated eigenvalues against N, which increases cross-sectionally. The observed jumps in the eigenvalue sequence reflect both the relatively small sample size and the fact that countries are not independent and identically distributed units. In particular, economies with greater weight in global trade exert a disproportionately large influence on overall variance. This explains the two marked jumps around observations 11 and 41, corresponding respectively to Germany and the United States. We observe that roughly three eigenvalues grow at a faster rate, while the remaining ones increase much more slowly, consistent with non-pervasive (idiosyncratic) behavior.



(a) Scree plot of variance share by factor

(b) Eigenvalues against N (pervasiveness check)

Figure 1: Diagnostic plots for factor selection

The eigenvalue–ratio rule favors a very parsimonious specification ($r = 1$), while the scree plot and the pervasiveness check both point to points to $r = 3$. We therefore proceed with an FNAR model with three factors.

The three factors account for 57% of total variation with the first factor explaining 46.4% of the variance, the second 6.0%, and the third 4.6%. This choice, although reflecting a careful balance among the various selection criteria, does not alter the fact that the first principal component remains the dominant source of variability. The second and third components, while capturing more variation than the remaining ones, still account for only a marginal share of the total variance and can thus be considered partly idiosyncratic.

3.2.2 Interpretation of Network Factors

Factor interpretation is a crucial component of the analysis, as it ensures that the results remain economically meaningful. We employ a heat map in which each panel reports the estimated network factors averaged over the period 2000–2014. Green cells represent positive entries, while red cells indicate negative ones; the darker the colour, the greater the weight of cross-country trade in the target country's total trade. This pattern is exemplified by the USA column, which displays strong trade linkages with Mexico and Canada.

The heat maps also highlight the strong presence of Germany across all three factors. This reflects the structure of our dataset, which is composed of roughly 60% European countries, thereby increasing Germany's relative weight in the total trade represented in our sample.

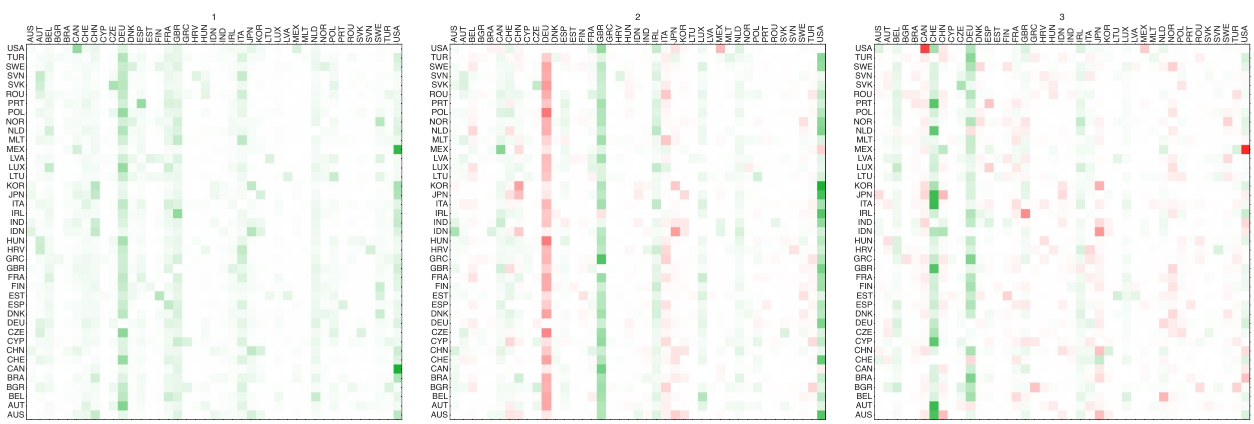


Figure 2: Average country weights on the selected three network factors over the period 2000–2014

The scatter plots below illustrate the loadings of our initial sectors on the factors found. This is essential for interpretation as it gives us the opportunity to classify by which sectors or sector category the factors are charactarised.

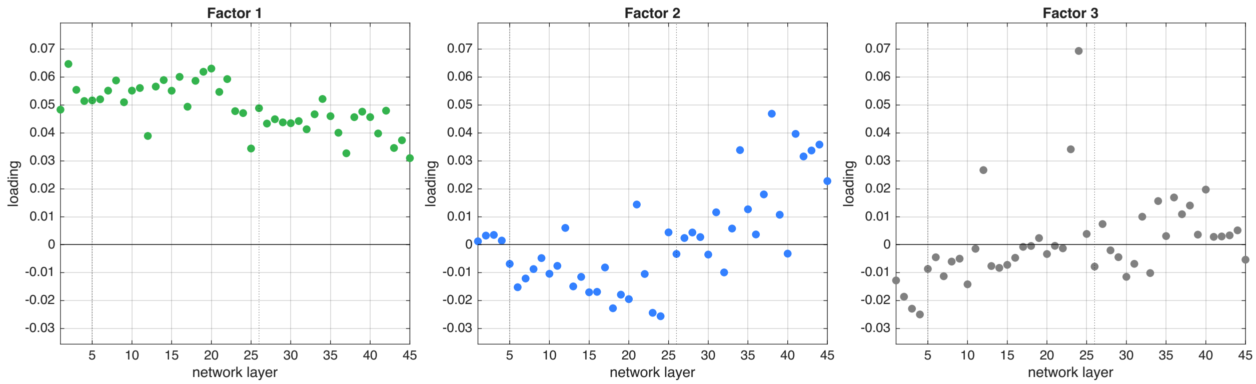


Figure 3: Factor loadings across network layers for the selected three network factors

The factors can be interpreted as follows. The first factor represents the overall level of international trade, which can be broadly associated with globalisation and shows a bias towards Europe. Loadings are positive and relatively homogeneous across sectors, indicating a general scale effect in cross-border linkages. Countries with high scores, most notably the United States and Germany, occupy central positions within the trade network and are active across a wide range of sectors. This factor therefore captures the common trend in global trade interconnectedness.

The second factor represents a sectoral-composition margin contrasting tertiary activities with primary/secondary industries. Countries such as the United States and the United Kingdom score positively, reflecting strong engagement in tradable services, while Germany scores negatively, consistent with a manufacturing-intensive profile. This factor therefore captures systematic cross-country specialisation along the goods-services dimension.

The third factor reflects trade in essential goods and services, where exchange is often shaped by infrastructural interdependence and geographic proximity. High absolute loadings in pharmaceuticals, electricity, water supply, and related services suggest that these sectors form tightly connected networks driven by the need for reliable cross-border provision. Positive country scores for Switzerland, Germany, Ireland, Luxembourg, Belgium, and, to a lesser extent, the United States, indicate specialisation in these infrastructure-intensive activities, characterised by stronger regional linkages compared to the more globalised patterns captured by the first two factors.

It is important to note that the choice of three factors, although justified by pervasive eigenvalues and the screeplot implicate that the second and third especially represent more idiosyncratic components compared to the first. For this reason the interpretation is more blurred and less evident.

Overall, the retained factors jointly summarise (i) the common rise in cross-sector trade interconnectedness, (ii) cross-country specialisation between manufacturing and services, and (iii) a narrower axis of essential good and service trade.

3.3 Estimation and Interpretation of OLS and GLS coefficients

After retaining the three network factors listed above, we proceed by estimating the FNAR in Equation (1) with both OLS and GLS, where our y_t are the first differences of unemployment rates

reported in World Bank dataset. First differences as dependent variables are used to respect the stationarity requirement for the interpretation of the coefficients in the analysis.
Given the presence of the lagged variable on the right hand side, our estimation has to be performed on years 2001-2014 (since WIOT only contains data for 2000-2014).

Table 1 reports all the coefficients of the FNAR for OLS and GLS, i.e. the three coefficients related to network factors and the other two related to momentum effect and nodal effect respectively.

The betas on factors in the case of OLS are significant for all the terms while using GLS the coefficient of the last one is not. This confirms our belief on the strong influence of the first factor specifically, as well as the weaker effect from the last one.

Table 1: FNAR coefficient estimates and standard errors.

	$\hat{\theta}^{\text{OLS}}$	$\hat{\theta}^{\text{GLS}}$
network effects		
$\hat{\beta}_1 N^{-1} \hat{F}_{1,t-1} y_{t-1}$	-0.7488^{***}	-0.4347^{***}
	(0.2798)	(0.1103)
$\hat{\beta}_2 N^{-1} \hat{F}_{2,t-1} y_{t-1}$	1.6275^{***}	0.4281^{**}
	(0.3944)	(0.1962)
$\hat{\beta}_3 N^{-1} \hat{F}_{3,t-1} y_{t-1}$	-1.3155^{***}	-0.2603
	(0.4980)	(0.2031)
momentum effect		
$\hat{\rho}\, y_{t-1}$	0.4965^{***}	0.4045^{***}
	(0.0465)	(0.0348)
nodal effect		
$\hat{\alpha}$	-0.0383	0.0418
	(0.1580)	(0.0774)

In the FNAR specification estimated over the full 2000-2014 sample, the coefficients $\hat{\beta}_j$ measure how changes in unemployment propagate across countries through the network structure. Each coefficient captures the responsiveness of current unemployment changes to past unemployment changes, conditional on a country's position along the corresponding network factor.

The first network coefficient $\hat{\beta}_1 < 0$ indicates that exposure to the first factor dampens the transmission of unemployment fluctuations: countries that are more interconnected along this globalisation related dimension tend to experience a smaller follow up change when unemployment shifted in the previous year, consistent with a stabilising propagation mechanism. The second coefficient $\hat{\beta}_2 > 0$ reflects an amplifying channel, whereby countries scoring higher on this factor, associated with service intensive and public administration sectors, tend to transmit past unemployment changes more strongly into the next period. The third coefficient $\hat{\beta}_3 < 0$ again suggests attenuation: integration in the essential-goods and utility-related sectors linked to this factor weakens the persistence of unemployment changes, providing a buffer against the propagation of shocks.

The momentum coefficient displays strong and highly significant persistence in unemployment changes, indicating that increases (or decreases) in one year tend to be followed by movements in the same direction the following year. Finally, the nodal term is statistically insignificant, implying that, once network propagation and momentum are accounted for, there is no residual systematic drift in unemployment changes at the aggregate level.

3.4 Pre-Post 2008 Comparison

Given that our dataset spans 2000-2014, it naturally includes both the 2008 global financial crisis and the subsequent European sovereign debt crisis. In the analysis below, we investigate the role played by global trade networks in shaping unemployment dynamics, with particular attention to differences between the pre- and post-2008 periods.

Although factor scores vary over time, the factor loadings do not. This makes it necessary to estimate the tensor-PCA separately for the two subperiods, 2000-2007 and 2008-2014, rather than relying on the loadings obtained from the full sample. Figure 4 shows that the sectoral loadings associated with the first three network factors remain remarkably similar across the two periods, indicating that the underlying network structure is stable over time.

To assess how these network factors influence unemployment before and after the crisis, we estimate two separate VAR(1) regressions for the two sub-periods. Given the limited sample size,particularly the small time dimension T, we restrict attention to OLS estimation without heteroskedasticity-robust standard errors.

Table 2: FNAR coefficient estimates before and after 2008.

	2000–2007	**2008–2014**
network effects		
$\hat{\beta}_1 N^{-1} \hat{F}_{1,t-1} y_{t-1}$	0.1996	-1.9289^{***}
	(0.3087)	(0.4103)
$\hat{\beta}_2 N^{-1} \hat{F}_{2,t-1} y_{t-1}$	0.5435	2.0246^{***}
	(0.6225)	(0.4953)
$\hat{\beta}_3 N^{-1} \hat{F}_{3,t-1} y_{t-1}$	0.2381	-3.6476^{***}
	(0.5598)	(0.7186)
momentum effect		
$\hat{\rho}\, y_{t-1}$	0.3886^{***}	0.6467^{***}
	(0.0539)	(0.0681)
nodal effect		
$\hat{\alpha}$	-0.1653^{**}	0.1358
	(0.0630)	(0.1107)

A clear structural shift emerges in the predictive role of network factors when comparing the pre- and post-2008 periods (Table 2). During 2000–2007, all three $\hat{\beta}$ coefficients are small and statistically insignificant, indicating that changes in unemployment across countries were largely orthogonal to the underlying trade-network structure.

Year-to-year fluctuations were instead related to country specific dynamics and the unemployment change of the previous year. A statistically significant momentum coefficient of 0.3886 indicates modest persistence in unemployment changes: an increase (or decrease) in one year tends to be followed by a smaller increase (or decrease) in the following year.

Before 2008, the intercept is negative and statistically significant, meaning that, controlling for network structure and persistence, unemployment tended to decline on average each year. This aligns with the broad reductions in unemployment observed across Europe during the mid 2000s expansion.

After 2008, the network pattern changes sharply. In 2008-2014, all three network coefficients become large in magnitude and highly significant. The first factor ($\hat{\beta} = -1.9289$) acts as a balancing mechanism: higher exposure on this factor predicts an opposite-signed change in unemployment relative to the previous year, consistent with a stabilising role of the broad globalisation margin. The second factor ($\hat{\beta} = 2.0246$) amplifies fluctuations: strong service sector interconnectedness magnifies the propagation of unemployment changes, while more manufacturing-oriented countries experience a weaker predictive linkage. The third factor ($\hat{\beta} = -3.6476$) again mitigates propagation, suggesting that greater integration in essential goods and services provides a buffer against sustained unemployment accelerations.

The momentum coefficient rises substantially after 2008, reaching 0.65, which indicates that changes in unemployment become markedly more persistent: increases are more likely to be followed by further increases, and improvements tend to continue rather than mean-revert.

Overall, unemployment dynamics become far more network-driven and persistent in the aftermath of the 2008 crisis. Whereas pre-crisis changes in unemployment were largely idiosyncratic, the post-crisis period is characterised by strong predictive relationships with trade-network structures and significantly higher inertia in year-to-year unemployment movements.

4 Conclusion

This paper applies the Factor Network Autoregression of Barigozzi et al. (2025) to a rich international trade dataset in order to assess whether sectoral trade interdependencies help forecast unemployment dynamics across countries.
Starting from a four-dimensional `Country` × `Country` × `Sector` × `Time` tensor constructed from WIOT data, we extract latent network factors via tensor-based PCA and embed them in a FNAR specification for first differences of unemployment rates. The resulting model allows us to decompose unemployment dynamics into a network effect, a momentum effect, and a nodal effect capturing common shifts.

Our factor-selection strategy combines the eigenvalue-ratio criterion, the scree plot of explained variance, and a pervasiveness check based on the growth of eigenvalues with the cross-sectional dimension. While the eigenvalue–ratio rule on its own points to a very parsimonious specification with a single factor, both the shape of the spectrum and the divergence pattern of eigenvalues support the presence of three economically meaningful common components. These three factors jointly explain 57% of the total variation in the multilayer trade network, with the first factor alone accounting for 46.4%, and the second and third contributing 6.0% and 4.6%, respectively.

This share of explained variance is lower than the one reported in the empirical application of Barigozzi et al. (2025), and this difference already emerges at the level of the first principal component, whose contribution (46.4%) is below the leading factor in their study (68%). A natural explanation is that our setting is both more granular and more heterogeneous: we work with 45 relatively disaggregated sectors that are less clustered than broad aggregates, and with a larger set of 41 countries. Greater sectoral disaggregation spreads variability across many layers instead of concentrating it into a few tightly correlated blocks, while the larger cross-country dimension increases heterogeneity in trade patterns. Together, these features make the common component less dominant and leave a larger role for idiosyncratic fluctuations at the country-sector level. As a consequence, the third component, especially, is closer to the boundary between common and idiosyncratic variation and should be interpreted with correspondingly greater caution.

Despite this more diffuse factor structure, the empirical FNAR estimates for the full sample highlight economically interpretable channels. The first factor, which we interpret as a globalisation component capturing the overall intensity of cross-border trade, enters with a negative coefficient: stronger participation in the global trade network contrasts the direction of change in unemployment rates. This means that it tends to mitigate shocks, independently of their sign. The second factor is associated with a positive propagation on the change in unemployment rates, which makes it procyclical. The third factor, linked to essential goods and infrastructural services such as electricity, water, and pharmaceuticals, generally exhibits a negative (though weaker and less robust) association with unemployment, in line with the stabilising role of necessity-based sectors. Across specifications, the momentum term confirms substantial persistence in unemployment changes, while the nodal effect remains statistically not significant, suggesting the absence of a strong common drift in first-differenced unemployment rates over the sample period.

With regards to our second analysis, which performs two estimations on the period preceding 2008 and on the one comprising it, we find similar results. However, such outcomes differ in the level of significance. Indeed, while prior to the Financial crisis the factors do not exhibit significant coefficients, this does not hold for the more recent time frame.
Coefficients for 2008-2014 are highly significant, suggesting a strong relationship between unemployment changes and the identified components.

Taken together, our results suggest that trade-based network factors carry meaningful information for unemployment dynamics. A small number of latent trade components, extracted from a large and heterogeneous country–sector system, already delivers statistically significant and economically interpretable effects within the FNAR. The fact that the three retained factors explain only a moderate share of total network variance is consistent with the high level of sectoral and cross-country detail of our dataset: a substantial part of the remaining variation naturally reflects idiosyncratic shocks and local specialisation patterns, rather than a shortcoming of the model. In this sense, our empirical exercise confirms that the FNAR framework can be successfully applied to rich multilayer trade data while still isolating a clear network channel for unemployment.

There are several natural directions for future work. A first extension would be to cluster countries according to initial labour-market or structural characteristics (such as average unemployment levels, income per capita, or degree of openness) and re-estimate the FNAR within more homogeneous groups. This would make it possible to compare how similar trade shocks propagate in economies with different starting conditions. A second improvement could extend the sample beyond 2014 in order to study whether the role of trade-network factors for unemployment becomes stronger, weaker, or simply different as the structure of global value chains evolves over time. Finally, one could test whether the lower explanatory power of the factors in the full sample reflects their weaker ability to predict unemployment changes over 2001–2004.

Appendix

Table 3: Sectors Excluded from the Analysis

Sector Description
Repair and installation of machinery and equipment
Sewerage: waste collection, treatment and disposal activities; materials recovery
Postal and courier activities
Motion picture, video and television programme production, sound recording and music
Activities auxiliary to financial services and insurance activities
Architectural and engineering activities; technical testing and analysis
Scientific research and development
Advertising and market research
Other professional, scientific and technical activities; veterinary activities
Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use
Activities of extraterritorial organizations and bodies

Table 4: Countries Included in the Analysis

Code	**Code**	**Code**	**Code**	**Code**	**Code**	**Code**
AUS	AUT	BEL	BGR	BRA	CAN	CHE
CHN	CYP	CZE	DEU	DNK	ESP	EST
FIN	FRA	GBR	GRC	HRV	HUN	IDN
IND	IRL	ITA	JPN	KOR	LTU	LUX
LVA	MEX	MLT	NLD	NOR	POL	PRT
ROU	SVK	SVN	SWE	TUR	USA	

Table 5: Sectors Included in the Analysis

Category	Sector	Description
Primary	1	Crop and animal production, hunting and related service activities
	2	Forestry and logging
	3	Fishing and aquaculture
	4	Mining and quarrying
Secondary	5	Manufacture of food products, beverages and tobacco products
	6	Manufacture of textiles, wearing apparel and leather products
	7	Manufacture of wood and of products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials
	8	Manufacture of paper and paper products
	9	Printing and reproduction of recorded media
	10	Manufacture of coke and refined petroleum products
	11	Manufacture of chemicals and chemical products
	12	Manufacture of basic pharmaceutical products and pharmaceutical preparations
	13	Manufacture of rubber and plastic products
	14	Manufacture of other non-metallic mineral products
	15	Manufacture of basic metals
	16	Manufacture of fabricated metal products, except machinery and equipment
	17	Manufacture of computer, electronic and optical products
	18	Manufacture of electrical equipment
	19	Manufacture of machinery and equipment n.e.c.
	20	Manufacture of motor vehicles, trailers and semi-trailers
	21	Manufacture of other transport equipment
	22	Manufacture of furniture; other manufacturing
	23	Electricity, gas, steam and air conditioning supply
	24	Water collection, treatment and supply
	25	Construction
Tertiary – Trade & Transport	26	Wholesale and retail trade and repair of motor vehicles and motorcycles
	27	Wholesale trade, except of motor vehicles and motorcycles
	28	Retail trade, except of motor vehicles and motorcycles
	29	Land transport and transport via pipelines
	30	Water transport
	31	Air transport
	32	Warehousing and support activities for transportation

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Category	Sector	Description
Tertiary – Information & Finance	33	Publishing activities
	34	Telecommunications
	35	Computer programming, consultancy and related activities; information service activities
	36	Financial service activities, except insurance and pension funding
	37	Insurance, reinsurance and pension funding, except compulsory social security
Tertiary – Professional & Administrative	38	Real estate activities
	39	Legal and accounting activities
	40	Administrative and support service activities
Tertiary – Public & Social	41	Accommodation and food service activities
	42	Public administration and defence; compulsory social security
	43	Education
	44	Human health
	45	Other service activities

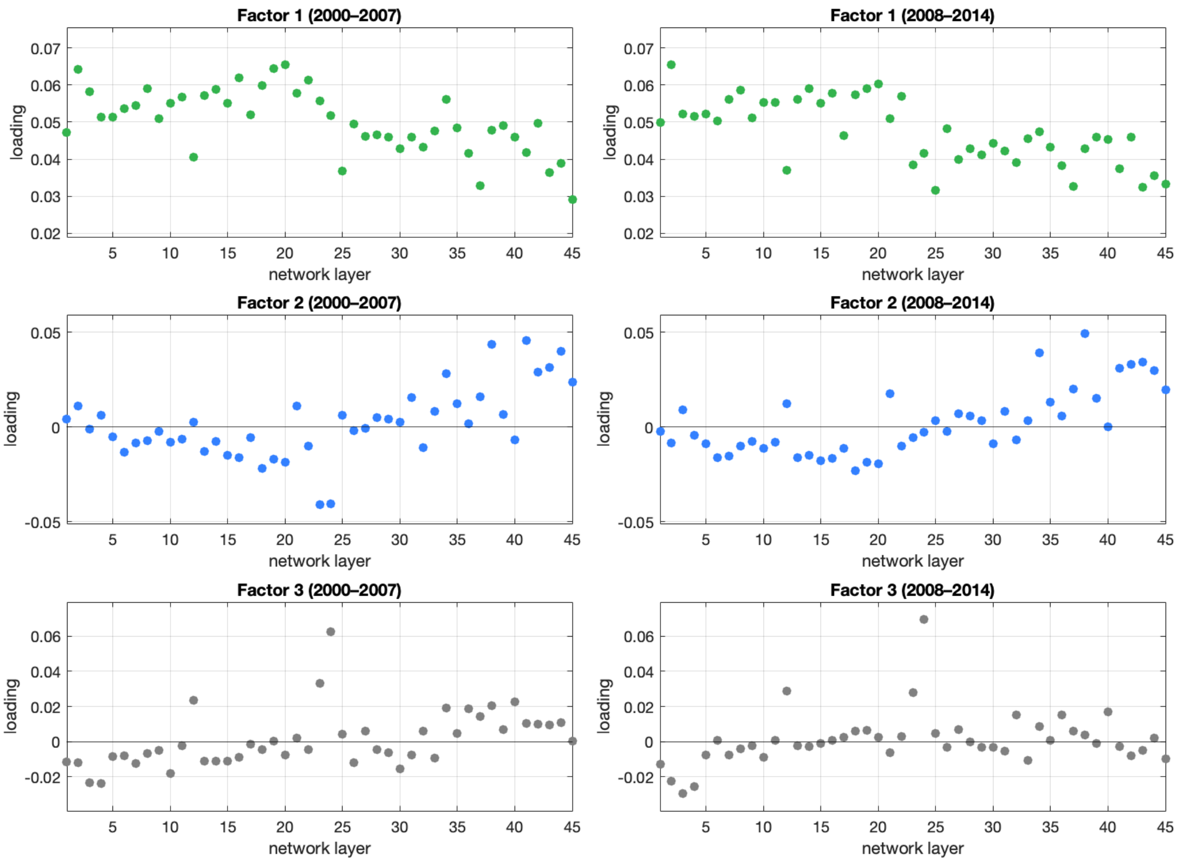

Figure 4: Factor loadings across network layers before and after 2008

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